

Experiment 19

Question

Perform Edge Detection using Sobel Matrix along XY axis.

Aim

To detect the edges of an image in both the X and Y directions using the Sobel operator in OpenCV.

Procedure

1. Import the OpenCV (cv2) library.
2. Read the input image in grayscale mode.
3. Apply the Sobel operator along both X and Y directions.
4. Display the original and edge-detected images.
5. Save and Display output image

Code

```
import cv2

# Read image
img = cv2.imread("dog.jpg", 0)

# Apply Sobel along X and Y
sobelxy = cv2.Sobel(img, cv2.CV_64F, 1, 1, ksize=3)

# Display images
cv2.imshow("Original Image", img)
cv2.imshow("Sobel XY", sobelxy)

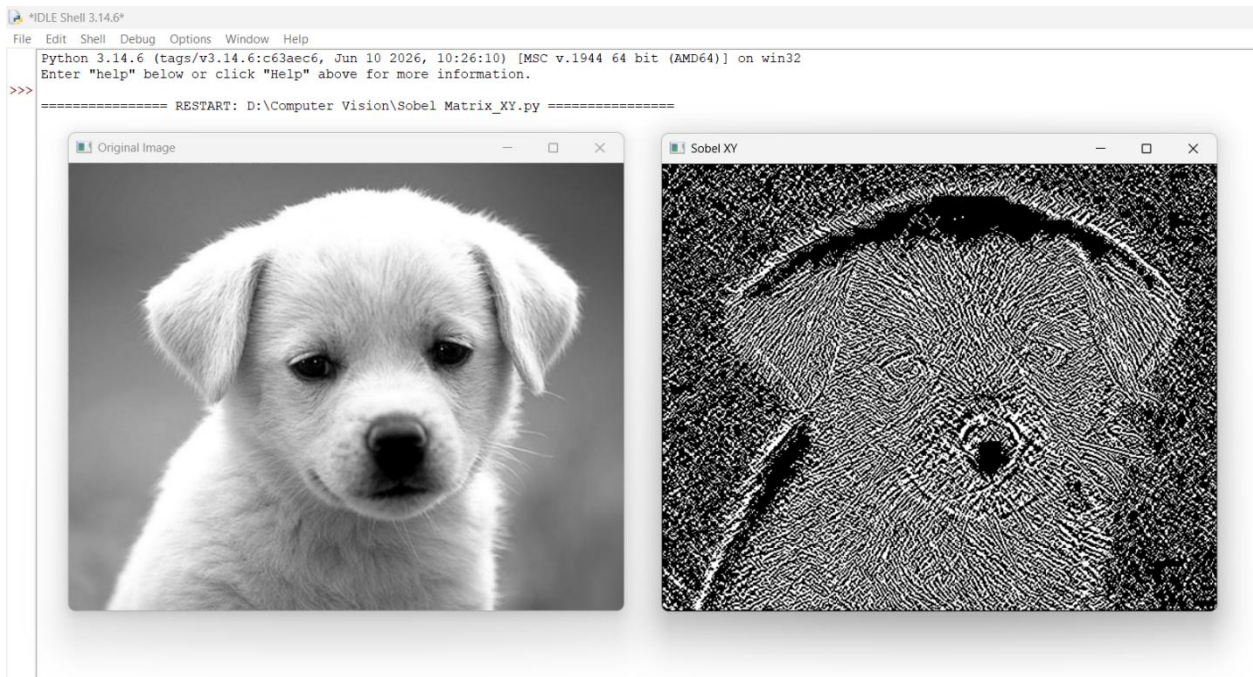
# Save output
cv2.imwrite("sobelxy_dog.jpg", sobelxy)
```

Input

Input Image: dog.jpg

Output

- Original Image
- Sobel XY Edge Image



Result

The edges of the input image were successfully detected in both the X and Y directions using Sobel operator.